



# Diabetes mellitus and functional limitations among older adults: Evidence from a large, representative Ghanaian aging study

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## ARTICLE INFO

### Keywords:

Diabetes  
Older adults  
Physical function  
Epidemiology  
Physical activity  
Sub-Saharan Africa

## ABSTRACT

**Objectives:** Literature linking diabetes mellitus (DM) to functional status is limited in low- and middle-income countries. Importantly, factors influencing this association are even less understood. This study aims to examine the association of DM with functional limitations (FL) in older adults and to identify potential factors influencing this association.

**Methods:** In a cross-sectional analysis, we examined the association between DM and basic and instrumental activities of daily living-related FL in 1201 adults aged  $\geq 50$  years from the Aging, Health, Psychological Well-being, and Health-seeking Behavior Study. DM was defined as a self-report of physician diagnosis. The associations were assessed using hierarchical regression estimates and bootstrapping technique via the Hayes PROCESS macro program.

**Results:** The prevalence of DM and FL was 10.1% and 36.1%, respectively, with OR = 2.50 (95%CI = 1.59–3.92) after accounting for sociodemographic factors, smoking, alcohol use, self-rated health, loneliness, and sleep quality. After full adjustment, polytomous regressions showed that the association of DM with FL increased with the number of FL (i.e., OR = 1.60 for 1–2, OR = 1.88 for 3–5, and OR = 2.00 for  $>5$  FL compared with no FL). However, this association was attenuated after controlling for physical activity (OR = 2.06, 95%CI = 1.28–3.31), hypertension (OR = 1.87, 95%CI = 1.14–2.99), stroke (OR = 1.82, 95%CI = 1.20–2.93), and pain facets (OR = 1.80, 95%CI = 1.04–3.02). PA thus mediated 40.39% of the DM-FL association.

**Conclusions:** In this representative study, older adults with DM showed higher odds for FL, and this association was partially explained by physical activity and health variables. Investing in a holistic management approach might be helpful for public health planning efforts to address DM-induced FL in old age.

## 1. Introduction

Global populations are increasingly growing, and for many older

people, age-related functional limitations (FL) remain a serious concern regarding access to and costs of healthcare [1,2]. FL denotes the difficulty or deterioration in performing basal (e.g., bathing or using the

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<https://doi.org/10.1016/j.jpsychores.2023.111481>

Received 26 October 2022; Received in revised form 4 August 2023; Accepted 29 August 2023

Available online 30 August 2023

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toilet) and instrumental activities of daily living (IADL, e.g., using a telephone or taking medication) [1]. Epidemiologic and prospective studies have linked FL in later life to depression [3], chronic conditions [4], poor quality of life [5], and increased risk of death [6], which in turn increase admission to nursing or old age home [7], and leads to economic burden on families and society. It is thus crucial to identify the risk factors of FL, particularly in low- and middle-income countries (LMICs) where the management of age-related FL may be fragmented.

Diabetes mellitus (DM) is a severe systemic metabolic disorder with increasing global prevalence, particularly among older persons. The International Diabetes Federation [8] estimates that 3 in 4 adults with diabetes reside in LMICs, and about 54% of people living with diabetes in the African Region are undiagnosed. Several prospective studies have shown a positive relationship between DM and FL among older adults [9,10]. DM may contribute to FL via an increase in health-related complications, including vision impairment [11], nephropathy, neuropathy, and higher risk for infections [12], lower extremity amputations [13], polypharmacy [14], underweight due to dieting pattern [15], as well as increased risk of hospitalization [16]. A systematic review and meta-analysis of 26 studies found that diabetes was independently associated with increased risk of activities of daily living (ADL) difficulty (OR = 1.82, 95%CI = 1.63–2.04), mobility decline (OR = 1.71, 95%CI = 1.53–1.91), and IADL impairment (OR = 1.65, 95%CI = 1.55–1.74) [17]. Among 729 older women, Volpato et al.'s [18] longitudinal analysis in Baltimore revealed that women with diabetes had increased odds of mobility and ADL disabilities. Indeed, the DM-FL association could be stronger in sub-Saharan Africa (SSA) than in high-income countries, given that the management of DM and its complications could be suboptimal in SSA. However, evidence on the association of DM with functional performance, and functional limitations, specifically, in older adults, is limited in SSA.

Crucially, despite the pathophysiologic heterogeneity of DM, there is no data on the factors influencing the association of DM with FL. To our knowledge, only one study has reported the effect of DM on FL in Ghana and South Africa [19]. However, this study did not evaluate how the association could be influenced by PA, hypertension, stroke, and pain despite the key roles of these factors in the etiology and progression of DM and functional decline in later life [25,47]. Recent literature has linked physical activity (PA) and health-related factors to DM [17,20]. A systematic review and meta-analysis of 17 studies in Ghana found that physical inactivity was a modifiable risk factor for DM [21]. For example, 150 min a week of moderate-to-vigorous PA was associated with a 27% reduced risk of DM [22]. Lin et al. [23] noted that patients with comorbidity profiles such as stroke and hypertension presented DM but were less likely to have diabetes-related face-to-face visits. Patients with DM and atypical symptoms are more likely to be diagnosed with angina pectoris and pain than those without DM [24]. Since DM has complex interrelationships with many factors, more data are needed to understand the factors influencing the DM-FL association.

Therefore, this population-based study aimed to examine the association of DM with FL in older adults in Ghana and to evaluate the factors influencing this association. Given that demographic aging and the associated chronic/functional health burden are expected to be significant in SSA [25], our findings may be crucial for public health planning.

## 2. Methods

### 2.1. Study population

The present cross-sectional study used data from the Aging, Health, Psychological Well-being, and Health-seeking Behavior Study devised to examine the health and health-seeking behavior of adults aged 50 years or over during 2016–17 [26,27]. The study included six randomly selected rural/urban districts in Ghana using a multi-stage stratified cluster sampling procedure. The sample size for the study was

determined using the estimation model by the WHO, provided as follows:

$$N = \rho \times \frac{\left[ \left( \bar{U}_{\omega} \right)^2 \times \pi(1 - \pi) \right]}{\delta^2}$$

The  $\pi$  is the expected prevalence of older adults in Ghana, and  $\rho$  denotes the design effect. We also presumed a 5% margin of error, 95% confidence interval (CI), 1.5 design effect, 5% type 1 error, 15% type 2 error, and 50% default prevalence of the target population. Given  $\pi = 0.5$  and  $\bar{U}_{\omega} = 1.96$ , we obtained the minimum sample of 901. Considering the losses in participation, we included an oversampling of 38%. This achieved a sample of 1247 for this study, and the model reached a statistical power of 85% and a 5% (two-sided) significance level to detect an odds ratio of  $\geq 2$ . Moreover, 28(2.3%) of the participants declined to participate, 15(1.2%) completed questionnaires had missing essential data, and 3(0.2%) contained substantial outliers in relation to the ages of the respondents reported (i.e., 21, 14, and 210 years). Therefore, the analysis involved 1201 older persons with complete data (Fig. 1). Trained research assistants conducted face-to-face interviews for data with a 96% response rate.

### 2.2. Ethical considerations

The Committee on Human Research, Publications & Ethics (CHRPE), School of Medical Sciences, Kwame Nkrumah University of Science and Technology, and Komfo Anokye Teaching Hospital, Kumasi, Ghana, approved the study protocol (Ref: CHRPE/AP/507/16). Study participants also completed written informed consent.

### 2.3. Functional limitations (FL)

FL (outcome) was evaluated with self-reported difficulties in conducting seven activities in relation to ADL and IADL using the Medical Outcomes Study Short Form-36 (MOS SF-36) Scale [28]. The items considered the extent to which older adults undertook ... 1) vigorous activities (e.g., weeding, lifting heavy objects, etc.), 2) moderate activities (e.g., moving a table/chair, washing, etc.), 3) lifting/carrying groceries, 4) climbing about several flights of stairs, 5) bending/kneeling/stooping, 6) walking more than one kilometer, and 7) bathing/dressing. Each item was scaled on a 4-point scale, from not limited at all = 1 to much limited = 4. We modeled FL in several ways to test the findings' robustness. First, we calculated the FL index (range: 7–28); higher scores indicated higher levels of FL (Cronbach alpha = 0.871). Second, the FL score was dichotomized into no FL = 0 and any FL = 1. Third, we classified the FL score into 0, 1–2, 3–5, and  $> 5$  for analytical purposes.

### 2.4. Diabetes mellitus (DM)

DM (exposure) was assessed based on the WHO Study on Global Aging and Adult Health (SAGE) INDIVIDUAL Questionnaire. In accordance with other large cohort studies [29], the participants were asked: 1) Have you ever been diagnosed by a doctor with DM (high blood sugar)? 2) Have you been taking insulin or other blood sugar-lowering medications...? The responses were gauged on a no/yes scale. Those who responded yes = 1 to any of the items were deemed to have DM.

### 2.5. Covariates

Our analyses controlled for other variables selected based on previous studies [13,26]. Socio-demographic factors included age (years), gender (female = 1/male = 2), residential type (rural = 1/urban = 2), marital status (married = 1/not married = 2) and educational status (primary-level/never = 1, secondary-level = 2, tertiary-level = 3). Current smoking status was assessed using the question, "Do you

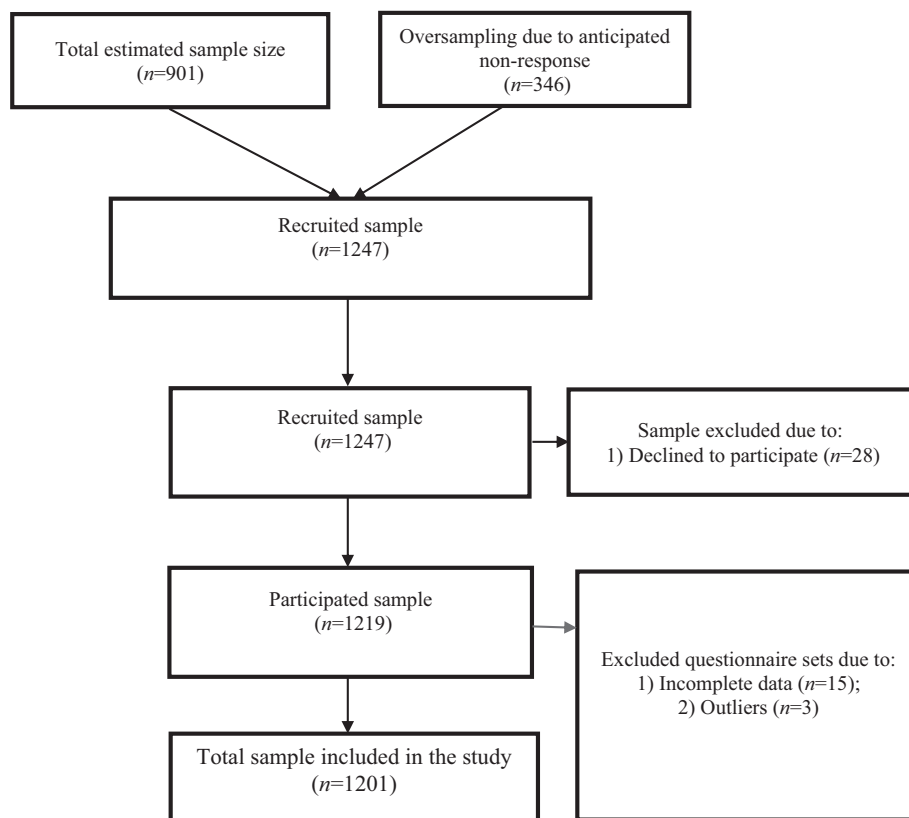


Fig. 1. Flow chart of the selection of study participants.

currently use any tobacco products such as Tawa, cigarettes, chewing tobacco, or snuff power? with a no=1/yes=2 response scale". *Current alcohol use* was assessed by asking the participants to indicate on a no = 1/yes = 2 response scale if they consumed any drink containing alcohol, such as beer, hard wine, or spirit, over the past 30 days. *Self-rated health* was assessed by asking the respondents to rate their overall health from the MOS SF-36 scale of general health [28] with a 5-Likert scale from excellent = 1 to poor = 5. Higher scores indicated higher levels of poor outcomes.

*Physical activity (PA)* was assessed with the International PA Questionnaire short form (IPAQ-SF) [30] validated in the older African population with good reliability and validity [31]. The tool measured the past week's dimensions of PA intensity (low, moderate, and vigorous) and asked the number of days that a participant was involved in various activities. E.g., "During the last seven days, on how many days did you walk for at least 10 minutes at a time"? A continuous scale was generated as activity events per week, with higher scores indicating higher levels of PA.

*Hypertension and stroke* were measured in accordance with a prior large study, and respondents were asked to self-report whether they had been diagnosed with hypotension and stroke by a doctor [29]. Responses were recorded as a no = 0/yes = 1. We assessed *pain severity* with a cross-culturally validated single-item, "How much bodily pain have you had during the past 4 weeks?" with response options: none = 1 to very severe = 5 [28]. We generated a continuous score ranging from 1 to 5, with higher scores indicating higher levels of pain severity.

*Pain interference* was assessed using a single-item evaluation of the impact of pain on specific daily activities using a 1–5 rating scale with established reliability and validity: "During the past 4 weeks, how much did pain interfere with your normal work?" [28]. The responses ranged from not at all = 1 to extreme = 5; higher scores reflected higher levels of pain interference. *Sleep quality* was measured with two items: "Overall, in the last 30 days, 1) how much of a problem did you have

with sleeping (e.g., falling asleep, waking up frequently during the night, or waking up too early in the morning)? and 2) how much of a problem did you have due to not feeling rested and refreshed during the day (for example, feeling tired, not having energy)?" [32]. Each item had five-point options from *none* = 1 to *extreme* = 5. Previous studies have used these items. A total score ranged from 2 to 10, with increasing scores indicating poor sleep quality (Cronbach's alpha = 0.830 in this study).

*Loneliness* was assessed with the University of California, Los Angeles 3-item loneliness scale: "How often do you (a) lack companionship?" (b) "Feel left out?" and (c) "Feel isolated?" with a 3-point scale: hardly ever/never, some of the time/sometimes, and often/always [33]. An overall score ranged from 3 to 9, with higher scores suggesting higher levels of loneliness (Cronbach's alpha = 0.810 in this study).

## 2.6. Statistical analysis

The statistical analyses were performed using SPSS version 25.0 (IBM Inc., Armonk, NY, USA). Frequency statistics were performed to characterize the sample. Data are presented as mean (standard deviation) for continuous variables and frequency counts (%) for categorical variables. The characteristics of persons with and without DM were compared using the independent sample *t*-tests for continuous variables or  $\chi^2$  tests for categorical variables. The normality of variable distribution was diagnosed with a Kolmogorov–Smirnov test with Lilliefors correction, and the homogeneity of variances was checked with Levene's test.

Multivariable data analytical methods were employed. The association of DM with FL was evaluated using logistic regression models with FL as the dichotomous outcome. We applied a hierarchical approach to assess the influence of DM-FL association by covariates. Model 1 adjusted for sociodemographics, smoking, alcohol consumption, and self-rated health. Model 2 added PA, while Model 3 included

hypertension. Model 4 subsequently included stroke, and Model 5 added pain characteristics. In sensitivity analysis, we treated FL as a fourth-level outcome (i.e., 0, 1–2, 3–5, and > 5) and evaluated the DM-FL association using multivariable polytomous logistic regressions with 0 FL as the reference. We reported estimates as odds ratio (OR) and 95% confidence interval (CI). We further deemed FL as a continuous index, and multiple OLS regressions assessed the association of DM with FL score. We further explored the potential mediating or confounding effect of PA in the association between DM and FL using bootstrapping mediation analysis from the Hayes PROCESS macro program. The unstandardized regression coefficients and standard errors were reported. In all regressions, multicollinearity was tested using the variance inflation factor (VIF). The largest variance found was 1.36, suggesting an absence of multicollinearity. All tests were two-tailed, and alpha was set at  $p < .05$  to determine statistical significance.

### 3. Results

#### 3.1. Descriptive and bivariate associations

The analytic sample consisted of 1201 adults aged  $\geq 50$  years with a mean age of 66.1 (SD = 11.9) years, and 63.3% were females. The majority were urban dwellers (55.0%), not married (56.6%), and attained up to primary school education (86.2%). Table 1 shows further details. The prevalence of FL and DM was 36.1% and 10.1%, respectively. Participants with DM were more likely to be uneducated ( $\chi^2 [2, N = 1201] = 8.564, p = .014$ ), non-alcohol users ( $\chi^2 [1, N = 1201] = 7.294, p = .007$ ), had lower levels of PA ( $t [1175, N = 1177] = 4.511, p < .001$ ), worse SRH ( $t [1199, N = 1201] = -4.411, p < .001$ ), and more

likely to have comorbidities ( $t [1199, N = 1201] = -6.700, p < .001$ ). Those with DM reported more FL than the DM-free adults ( $\chi^2 [1, N = 1201] = 31.837, p < .001$ ). The prevalence of DM increased with increasing levels of FL (Fig. 2). For example, the proportion of DM was 6% among those without FL but increased to 13% for those with 1–2, 16% for those with 3–5, and 23% for those with >5.

#### 3.2. Main regression and mediation analyses

Table 2 shows the multivariable analysis of the association of DM with FL. After adjusting for sociodemographic, smoking, alcohol use, loneliness, and sleep quality (Model 1), regressions showed that persons with DM were 2.50 times more likely to have FL compared with DM-free persons (OR = 2.50, 95%CI = 1.59–3.92). Accounting for PA in Model 2 substantially attenuated the significant DM-FL association (OR = 2.06, 95%CI = 1.28–3.31). Mediation analysis revealed that DM was significantly associated with lower levels of PA ( $\beta = -0.465, p < .001$ ) but higher levels of FL ( $\beta = 0.557, p < .01$ ). In addition, PA was related to lower PF levels ( $\beta = -0.483, p < .001$ ) (see Fig. 3). Importantly, PA significantly explained 40.39% of the association between DM and FL (see Table 3). With the additional adjustment for hypertension, the association of DM with FL remained significant and weakened (OR = 1.87, 95%CI = 1.14–2.99) (Model 3). In Model 4, we found a statistically significant association between stroke and FL, and its inclusion slightly attenuated the DM-FL association (OR = 1.82, 95%CI = 1.20–2.93). In the final model, pain severity and pain interference were significantly related to FL, and the DM-FL association waned negligibly (OR = 1.80, 95%CI = 1.04–3.02).

**Table 1**  
Demographic and health-related characteristics of the sample by DM status.

| Variable                   | Overall |        | No DM |        | DM   |        | Statistic | p-value |
|----------------------------|---------|--------|-------|--------|------|--------|-----------|---------|
|                            | N       | (%)    | N     | (%)    | N    | (%)    |           |         |
| Age (years) (M $\pm$ SD)   | 66.1    | (11.9) | 66.1  | (12.0) | 66.4 | (10.4) | -0.289    | 0.772   |
| Sex                        |         |        |       |        |      |        |           |         |
| Female                     | 760     | (63.3) | 683   | (63.2) | 77   | (63.6) | 0.007     | 0.932   |
| Male                       | 441     | (36.7) | 397   | (36.8) | 44   | (36.4) |           |         |
| Settlement type            |         |        |       |        |      |        | 0.784     | 0.376   |
| Rural                      | 540     | (45.0) | 481   | (44.5) | 59   | (48.8) |           |         |
| Urban                      | 661     | (55.0) | 599   | (55.5) | 62   | (51.2) |           |         |
| Marital status             |         |        |       |        |      |        | 1.136     | 0.287   |
| Married                    | 521     | (43.4) | 463   | (42.9) | 58   | (47.9) |           |         |
| Not married                | 680     | (56.6) | 617   | (57.1) | 63   | (52.1) |           |         |
| Level of education         |         |        |       |        |      |        | 8.564     | 0.014   |
| Never                      | 1035    | (86.2) | 937   | (86.8) | 98   | (81.0) |           |         |
| Basic-level                | 104     | (8.7)  | 94    | (8.7)  | 10   | (8.3)  |           |         |
| Secondary or higher        | 62      | (5.2)  | 49    | (4.5)  | 13   | (10.7) |           |         |
| Current smoking            |         |        |       |        |      |        | 1.975     | 0.160   |
| No                         | 1068    | (88.9) | 965   | (89.4) | 103  | (85.1) |           |         |
| Yes                        | 133     | (11.1) | 115   | (10.6) | 18   | (14.9) |           |         |
| Current alcohol intake     |         |        |       |        |      |        | 7.294     | 0.007   |
| No                         | 823     | (68.5) | 727   | (67.3) | 96   | (79.3) |           |         |
| Yes                        | 378     | (31.5) | 353   | (32.7) | 25   | (20.7) |           |         |
| PA (M $\pm$ SD)            | 3.0     | (1.5)  | 3.1   | (1.5)  | 2.4  | (1.4)  | 4.511     | <0.001  |
| SRH (M $\pm$ SD)           | 3.4     | (0.8)  | 3.4   | (0.8)  | 3.8  | (0.8)  | -4.411    | <0.001  |
| Sleep quality (M $\pm$ SD) | 4.9     | (1.9)  | 4.8   | (1.9)  | 5.4  | (1.9)  | -2.88     | 0.004   |
| Loneliness (M $\pm$ SD)    | 5.3     | (3.9)  | 1.7   | (0.8)  | 2.0  | (0.9)  | -3.398    | 0.001   |
| Pain severity (M $\pm$ SD) | 3.2     | (1.1)  | 3.2   | (1.1)  | 3.6  | (1.0)  | -4.702    | <0.001  |
| Pain interference          | 3.0     | (1.3)  | 3.0   | (1.3)  | 3.5  | (1.2)  | -3.915    | <0.001  |
| Hypertension               |         |        |       |        |      |        | 42.697    | <0.001  |
| No                         | 771     | (64.2) | 726   | (67.2) | 45   | (37.2) |           |         |
| Yes                        | 430     | (35.8) | 354   | (32.8) | 76   | (62.8) |           |         |
| Stroke                     |         |        |       |        |      |        | 46.446    | <0.001  |
| No                         | 1152    | (95.9) | 1050  | (97.2) | 102  | (84.3) |           |         |
| Yes                        | 49      | (4.1)  | 30    | (2.8)  | 19   | (15.7) |           |         |
| FL                         |         |        |       |        |      |        | 31.837    | <0.001  |
| Not limited                | 767     | 63.9   | 718   | (66.5) | 49   | (40.5) |           |         |
| Any limitations            | 434     | 36.1   | 362   | (33.5) | 72   | (59.5) |           |         |

**Note:** N—frequency; M—mean; SD—standard deviation; FL—functional limitations; DM—diabetes mellitus; PA—physical activity; SRH—self-rated health. Test statistics and p-values are based on either  $\chi^2$  tests or Student's *t*-tests.

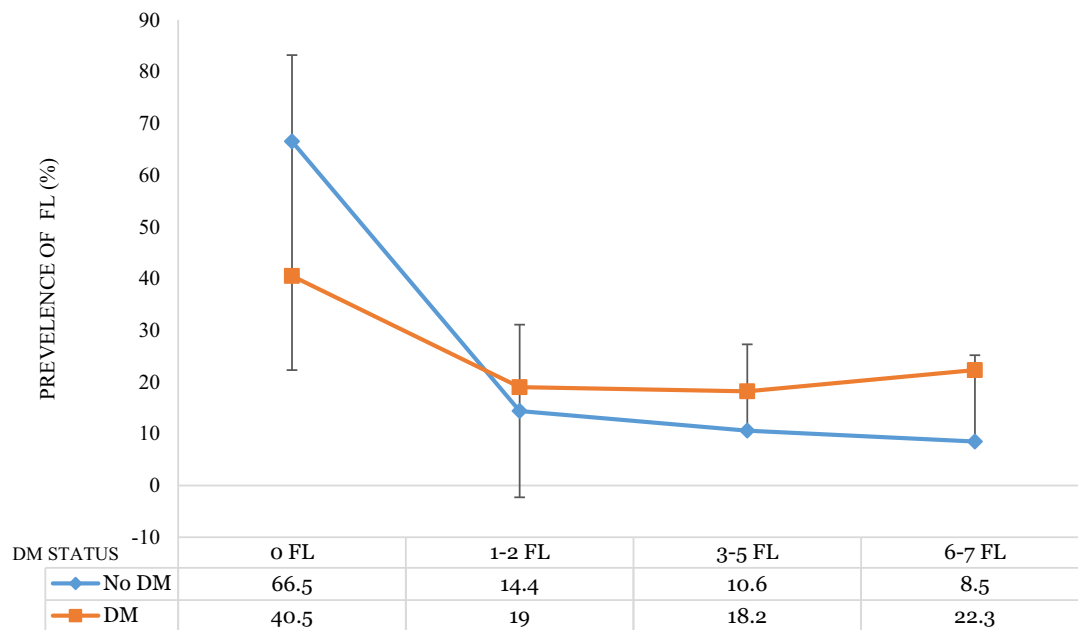


Fig. 2. Prevalence of functional limitations (FL) by type 2 diabetes mellitus (DM) status.

Table 2

Hierarchical multiple logistic regression models estimating the association between DM and FL.

| Variable                       | Model 1 |                | Model 2 |                | Model 3 |                | Model 4 |                | Model 5 |                |
|--------------------------------|---------|----------------|---------|----------------|---------|----------------|---------|----------------|---------|----------------|
|                                | OR      | (95% CI)       | OR      | (95% CI)       | OR      | (95% CI)       | OR      | (95% CI)       | OR      | (95% CI)       |
| Age (years)                    | 1.05    | (1.04–1.06)*** | 1.04    | (1.02–1.05)*** | 1.03    | (1.02–1.05)*** | 1.03    | (1.02–1.05)*** | 1.04    | (1.02–1.05)*** |
| Sex (ref: Female)              |         |                |         |                |         |                |         |                |         |                |
| Male                           | 1.65    | (1.15–2.36)**  | 1.43    | (0.97–2.10)    | 1.37    | (0.93–2.02)    | 1.38    | (0.93–2.03)    | 1.25    | (0.82–1.89)    |
| Residential type (ref: Rural)  |         |                |         |                |         |                |         |                |         |                |
| Urban                          | 0.91    | (0.68–1.21)    | 0.82    | (0.61–1.11)    | 0.77    | (0.57–1.05)    | 0.76    | (0.56–1.04)    | 0.76    | (0.54–1.07)    |
| Marital status (ref: Married)  |         |                |         |                |         |                |         |                |         |                |
| Not married                    | 1.35    | (0.98–1.87)    | 1.19    | (0.85–1.67)    | 1.16    | (0.82–1.63)    | 1.15    | (0.81–1.62)    | 1.17    | (0.81–1.69)    |
| Education (ref: Never)         |         |                |         |                |         |                |         |                |         |                |
| Basic-level                    | 2.36    | (1.39–4.00)*** | 2.29    | (1.31–3.99)**  | 2.36    | (1.34–4.14)**  | 2.34    | (1.33–4.11)**  | 3.20    | (1.75–5.85)*** |
| Secondary or higher            | 1.06    | (0.54–2.08)    | 0.99    | (0.50–1.96)    | 0.92    | (0.46–1.85)    | 0.92    | (0.46–1.85)    | 0.87    | (0.42–1.80)    |
| Smoking                        | 0.98    | (0.58–1.65)    | 0.89    | (0.51–1.57)    | 0.94    | (0.54–1.67)    | 0.94    | (0.53–1.66)    | 0.87    | (0.46–1.64)    |
| Alcohol intake                 | 0.83    | (0.59–1.17)    | 0.76    | (0.53–1.09)    | 0.76    | (0.53–1.09)    | 0.76    | (0.53–1.09)    | 0.68    | (0.46–1.01)    |
| SRH                            | 2.73    | (2.18–3.41)*** | 2.70    | (2.13–3.43)*** | 2.74    | (2.15–3.49)*** | 2.73    | (2.14–3.48)*** | 1.94    | (1.47–2.57)*** |
| Loneliness                     | 1.41    | (1.17–1.69)*** | 1.35    | (1.11–1.63)*** | 1.35    | (1.12–1.64)**  | 1.35    | (1.12–1.63)**  | 1.35    | (1.09–1.66)**  |
| Sleep quality                  | 1.12    | (1.04–1.22)**  | 1.08    | (0.70–1.18)    | 1.09    | (0.10–1.19)    | 1.09    | (1.00–1.19)*   | 1.10    | (1.00–1.21)    |
| PA                             |         |                | 0.62    | (0.55–0.69)*** | 0.63    | (0.56–0.70)*** | 0.63    | (0.56–0.70)*** | 0.61    | (0.54–0.69)*** |
| Hypertension                   |         |                |         |                | 1.67    | (1.23–2.28)*** | 1.66    | (1.22–2.27)*** | 1.88    | (1.34–2.63)*** |
| Stroke                         |         |                |         |                |         |                | 1.30    | (1.20–2.91)*   | 0.83    | (0.37–1.88)    |
| Pain severity                  |         |                |         |                |         |                |         |                | 1.48    | (1.17–1.86)*** |
| Pain interference              |         |                |         |                |         |                |         |                | 1.41    | (1.19–1.67)*** |
| DM                             | 2.50    | (1.59–3.92)*** | 2.06    | (1.28–3.31)*** | 1.87    | (1.14–2.99)**  | 1.82    | (1.20–2.93)*   | 1.80    | (1.04–3.02)*   |
| <b>Model Information</b>       |         |                |         |                |         |                |         |                |         |                |
| –2 Log-likelihood              |         | 1200.806       |         | 1096.857       |         | 1086.274       |         | 1085.840       |         | 945.151        |
| Nagelkerke R <sup>2</sup>      |         | 0.364          |         | 0.424          |         | 0.433          |         | 0.433          |         | 0.494          |
| R <sup>2</sup> Δ               |         | 0.014          |         | 0.060          |         | 0.009          |         | 0.000          |         | 0.061          |
| Hosmer-Lemeshow χ <sup>2</sup> |         | 13.732 (0.089) |         | 0.894 (0.065)  |         | 39.350 (0.061) |         | 34.255 (0.057) |         | 17.741 (0.052) |

Note: OR—odds ratio; CI—confidence interval; DM—diabetes mellitus; FL—functional limitations; PA—physical activity; SRH—self-rated health.

Model 1 was adjusted for age, sex, residential type, marital status, education, smoking, alcohol intake, SRH, loneliness, and sleep quality.

Model 2 included Model 1 and physical activity.

Model 3 included Model 2 and hypertension.

Model 4 included Model 3 and stroke.

Model 5 included Model 4, pain intensity, and pain interference.

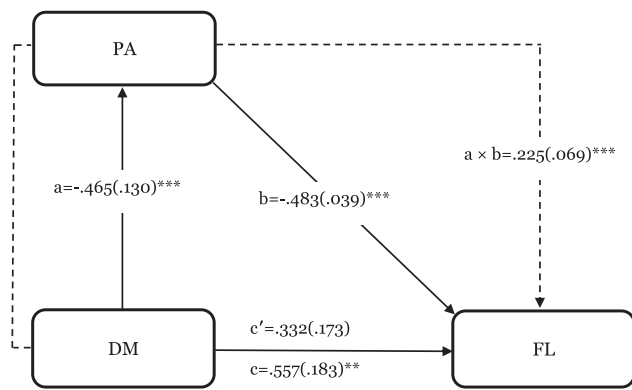
\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

### 3.3. Additional analyses

Table 4 presents the association between DM and the severity of FL risk. After the final adjustment (Model 4), polytomous regressions showed that the association was prominent with increasing severity of

FL. Compared with DM-free, those with DM were 60% more likely to present 1–2 (OR = 1.60, 95%CI = 1.11–2.84), 94% more likely to report 3–5 (OR = 1.94, 95%CI = 1.03–3.62), and approximately 100% more likely to have >5 (OR = 2.00, 95%CI = 1.34–3.96). Another sensitivity analysis considered the continuous FL score (see Table 5). After





**Fig. 3.** Bootstrapping mediation analysis of the effect of PA on the association of DM with FL. *Note:*  $a \times b$  = indirect effect of DM on FL;  $c'$  = direct effect of DM on FL;  $c$  = total effect of DM on FL; DM = diabetes mellitus; FL = functional limitations; PA = physical activity.

\*\*\* $p < 0.001$ ; \*\* $p < 0.01$ .

**Table 3**

The direct and indirect effects of DM on FL via PA.

| Path model                              |                 | $\beta$ | BootSE | Boots 95%CI  | Relative % mediated |
|-----------------------------------------|-----------------|---------|--------|--------------|---------------------|
| DM $\rightarrow$ PA<br>$\rightarrow$ FL | Direct effect   | 0.332   | 0.173  | −0.007–0.672 |                     |
|                                         | Indirect effect | 0.225   | 0.069  | 0.096–0.364  | 40.39               |
|                                         | Total effect    | 0.557   | 0.183  | 0.197–0.917  |                     |

**Note:**  $\beta$ —Unstandardized regression coefficients are reported; *BootSE*—Bootstrapping standard error, FL = functional limitations; DM = diabetes mellitus; PA = physical activity.

The model was adjusted for age, sex, residential type, marital status, level of education, smoking status, alcohol consumption, loneliness, self-rated health, and sleep quality.

Empirical 95% confidence interval does not overlap with zero.

adjustment for potential confounders, multiple OLS regressions found that the predicted FL score on a scale ranging from 7 to 28 is, on average, 0.37 ( $B = 0.37$ ,  $p < .001$ ) higher for those with DM than for those without DM.

## 4. Discussion

### 4.1. Principal findings

In this large, representative population-based study of older adults in Ghana, we found that DM was independently associated with higher odds of FL in a dose-response manner. Persons with DM had incremental

higher rates of FL than those without DM. The findings were similar across several FL definitions and analytic methods that made different methodological assumptions. Further, the positive DM-FL association was largely attenuated after adjusting for PA, hypertension, stroke, and pain. We further explored the potential mediating or confounding effect of PA in the DM-FL link. Bootstrapping analysis revealed that PA significantly explained approximately 40% of the association. Indeed, the overall prevalence of FL was high ( $\sim 36\%$ ), suggesting a substantial need to protect functional abilities through nuanced clinical treatment and management modalities for older adults, particularly in the SSA context. Our study markedly extends our current knowledge regarding an association between DM and FL in a neglected setting.

### 4.2. Interpretation of findings

The results of this study are consistent with our hypothesis and previous studies on the topic of DM and FL in old age. A prior population-based Health Survey in Finland found that DM was statistically significantly associated with greater handgrip strength decline among older men, but the analysis did not evaluate multifactorial biological pathways as potential influencing mechanisms [35]. Several studies have also suggested that DM may independently predict a higher risk of muscle dysfunction, physical frailty, mobility disability, and FL in older populations of diverse characteristics [13,17,36–39]. However, these studies largely included Western and Asian samples and did not consider the influence of PA, chronic physical conditions, and pain in the DM-FL association. The present study adds to the extant literature by using an innovative data set of older adults from the SSA context (where contemporary aging and the incidence of DM are predominantly telling) to evaluate the incremental effects of proximate factors in this association. Our data suggest a holistic approach in policy and public health efforts to effectively address physical dysfunction in old age.

Although the pathogenesis and biological mechanisms linking DM and FL are unclear, several plausible hypotheses could be proposed. Older persons with DM complications increasingly experience loss of muscle mass, mainly in the lower limbs [35,37], leading to underlying pathophysiological changes in sarcopenia, increased risks of falls, and frailty [37] which may reduce muscle strength [35]. Thus, impaired muscle performance may reduce gait speed in older adults [40]; severe hyperglycemia and insulin resistance may be associated with slower walking speed as peripheral neuropathy influences the effect of DM on walking performance [34]. Studies have shown that persons with DM lose muscle quality more rapidly and have poorer glycemic control than those without DM [41], which are known risk factors for FL. Older persons with DM are more likely to develop mental health complications, including depression and anxiety [42], which are important predictors of reduced gait speed and FL. Studies have shown that depression is related to low levels of physical activity, which, in turn, facilitates the onset and progression of FL, particularly in later life [41]. DM can

**Table 4**

Hierarchical polytomous logistic regression models, estimating the association between DM and the severity of FL.

| Model   | 1–2 FL |                | 3–5 FL |                | 6–7 FL |                | Model information       |                             |                   |
|---------|--------|----------------|--------|----------------|--------|----------------|-------------------------|-----------------------------|-------------------|
|         | OR     | (95% CI)       | OR     | (95% CI)       | OR     | (95% CI)       | Final −2 Log-likelihood | Pseudo Nagelkerke ( $R^2$ ) | Deviance $\chi^2$ |
| Model 1 | 2.36   | (1.35–4.07)**  | 3.31   | (1.86–5.90)*** | 5.41   | (3.10–9.37)*** | 1685.204                | 0.196                       | 1450.899***       |
| Model 2 | 2.11   | (1.18–3.70)*** | 2.78   | (1.53–5.04)*** | 2.78   | (1.43–5.42)*** | 1939.847                | 0.331                       | 1887.815**        |
| Model 3 | 1.80   | (1.01–3.20)**  | 2.31   | (1.24–4.31)**  | 2.35   | (1.15–4.80)*   | 1898.726                | 0.412                       | 1890.886          |
| Model 4 | 1.60   | (1.11–2.84)*   | 1.88   | (1.01–3.60)**  | 2.00   | (1.32–3.93)*   | 1884.675                | 0.429                       | 1882.718          |

**Note:** The reference category: No FL. OR—odds ratio; CI—confidence interval; DM—diabetes mellitus; FL—functional limitations.

**Model 1** was adjusted for age, sex, residential type, marital status, education, smoking, alcohol intake, SRH, loneliness, and sleep quality.

**Model 2** included Model 1 and physical activity.

**Model 3** included Model 2 and hypertension.

**Model 4** included Model 3 and stroke.

**Model 5** included Model 4, pain intensity, and pain interference.

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

**Table 5**

Hierarchical OLS regression models estimating the association between DM and FL composite index (range: 7–28).

| Model                    | Model 1   |             | Model 2   |             | Model 3   |             | Model 4   |             |
|--------------------------|-----------|-------------|-----------|-------------|-----------|-------------|-----------|-------------|
|                          | B         | (SE)        | B         | (SE)        | B         | (SE)        | B         | (SE)        |
| No DM                    | 0.00      | (0.00–0.00) | 0.00      | (0.00–0.00) | 0.00      | (0.00–0.00) | 0.00      | (0.00–0.00) |
| DM                       | 1.11      | (0.18)***   | 0.63      | (0.13)***   | 0.40      | (0.15)**    | 0.37      | (0.14)**    |
| <i>Model information</i> |           |             |           |             |           |             |           |             |
| Constant                 | –3.031    | (0.45)***   | 0.43      | (0.48)***   | –1.181    | (0.474)*    | –1.172    | (0.474)*    |
| F statistic ( $R^2$ )    | 28.822*** | (0.143)     | 42.013*** | (0.262)     | 52.745*** | (0.331)     | 50.809*** | (0.341)     |

**Note:** B—unstandardized regression coefficient; SE—robust standard error; OLS—ordinary least squares; DM—diabetes mellitus; FL—functional limitations.

**Model 1** was adjusted for age, sex, residential type, marital status, education, smoking, alcohol intake, SRH, loneliness, and sleep quality.

**Model 2** included Model 1 and physical activity.

**Model 3** included Model 2 and hypertension.

**Model 4** included Model 3 and stroke.

**Model 5** included Model 4, pain intensity, and pain interference.

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

increase the risk of other chronic conditions, such as cardiovascular diseases [44], peripheral neuropathy, osteoarthritis, and visual deficit [45], which are related to FL in older people. Moreover, anorexia and eating disorders due to changes in DM-induced dieting structure and patterns may lead to an underweight and sedentary lifestyle or less physical activity, which have been linked to poor functional performance in later life [15].

Crucially, our analysis evaluated the potential mediating or confounding role of PA in the association between DM and FL. We found that participants who engaged in PA were approximately 38% less likely to report FL. Indeed, controlling for PA substantially attenuated the association of DM with FL in this study. This finding is interesting and connotes the potential mediating or confounding effect of PA, explaining about 40% of the association of DM with FL. Thus, higher levels of DM can meaningfully reduce the levels and intensity of PA, particularly among aging adults, which, in part, may lead to increases in the severity of FL [19]. Conversely, FL may lead to reduced PA, which may, in turn, worsen DM. On the other hand, observational and population-based studies have indicated the relevance of PA in postponing or delaying the onset and progression of FL in older adults [46,47]. In a prospective study of adults with DM aged  $\geq 70$ , García Díaz et al. [48] found that the prevalence of frailty was reduced from 34.1% to 25%, and the percentage of FL also reduced from 26.2% to 21.4% after completing a 6-month exercise program. The advantages of aerobic exercise in older DM patients have been established. For example, walking decreases poor physical and muscular strength and reduces arterial stiffness in older adults [49], which are risk factors for FL. The association of physical inactivity with increased abdominal adiposity, leading to reduced muscle mass/strength, may predispose older adults to FL. Indeed, PA can enhance glycemic management by increasing insulin sensitivity, improving cardiovascular function, reducing blood pressure, and improving the lipid profile in persons with DM [48]. Ekelund et al. [51] noted in the InterAct case-cohort study that less PA increased the risk of DM and its proximate complications. Improvement in regular PA is critical to tackling the progression of FL, particularly among those with DM.

After controlling for hypertension, stroke, and pain, the association of DM with FL significantly attenuated but remained statistically significant. First, Our results showed that hypertension (OR = 1.65, 95%CI = 1.21–2.24) and stroke (OR = 1.32, 95%CI = 1.21–2.84) significantly increased the risk of FL and subsequently reduced the odds of DM on FL. This finding is consistent with a systematic review and meta-analysis of 66 studies suggesting a strong independent ischemic stroke-FL association [53]. The presence of chronic physical conditions such as hypertension and stroke with DM partly impacted the risk of FL among older adults. Many older adults, particularly in LMICs, more commonly present with hypertension and stroke, which mostly coexist with other health challenges [4,25]. Thus, several comorbidities and associated impairments that could propel FL are commonly prevalent in older

persons with DM. This suggests that the direct effect of DM on FL could be chiseled by the strong association of hypertension and stroke events with FL [54] these may work together and compete away the impact of DM on FL. This finding is, somewhat, in line with a systematic review of 50 studies reporting that higher blood pressure independently influenced cognitive performance in different domains among older adults [52]. The implication is that clinical and public health efforts to address DM-induced FL in old age should comprehensively subsume comorbidities. A significant contribution of the study involving a contemporary aging cohort is the addition of inclusion of pain serenity and subsequent interference as a confounder. The DM-FL association was weakened by pain severity and pain interference. Global studies have reported that DM is most strongly associated with pain experiences [24], while pain facets, in turn, relate strongly with FL among older adults [45; 55]. Indeed, pain results from DM and its complications. Diabetes and its complications, such as nephropathy and neuropathy of the lower limbs, are risk factors for intense pain and pain [11–13] interferences which may interfere with DM-FL association [12]. Given the recent increase in the burden of DM complications in old age in the LMICs, more efforts are needed to comprehensively manage DM and its complications, such as stroke, hypertension, and pain among older adults with diabetes.

#### 4.3. Strengths and limitation

This study contributes to understanding how PA, chronic diseases, and pain impact FL among older persons with DM. The key strength is that the analysis used a large and representative sample of community-dwelling older adults in a novel SSA context where the link between DM and FL is poorly understood. Indeed, a valid tool was used to quantify FL. However, our findings should be considered in the context of several limitations. DM was self-reported and our analysis did not consider the severity of DM and thus under-reporting and misclassification are likely. The cross-sectional design utilized in this study did not establish causal inferences. Future research would benefit from longitudinal data to validate these findings. The measurement of our key variables was based on retrospective self-reports and thus potentially introducing recall and social desirability bias into the findings. Future studies might consider an objective assessment of variables which may enhance the measures. One could add that the severity of DM in this study remains unknown. Our analysis did not adjust for psychological and mental health-related variables. Future research should consider key variables such as depression and anxiety in estimating the effect of DM and functional decline in later life.

#### 5. Conclusions

This population-based study showed that older adults with DM were more likely to have FL. However, a substantial portion of this association is explained by PA and marginally influenced by hypertension, stroke,

and pain characteristics. Since the trend in demographic aging is becoming unprecedented, particularly in LMICs, the incidence of DM may become more apparent, increasing the need for robust disability-related healthcare resources. Promoting PA and reducing hypertension, stroke, and chronic pain could be targets for improved DM-induced FL outcomes in old age. Public health practitioners should consider these factors in their clinical practice. Prospective studies are warranted to take care of reverse causation and to confirm these findings. In addition, future studies would benefit from further exploring and estimating additional potential mediators of the link between DM and functional limitations among older adults.

## Funding

This work was supported by Research Grant Council and Lingnan University, Hong Kong [grant numbers: RPG1129310], to Razak M. Gyasi (<https://www.ln.edu.hk/about-lu/introducinglingnan>). The funders had no role in the study design, data collection, and analysis, decision to publish, or preparation of the manuscript.

## CRediT authorship contribution statement

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## Declaration of Competing Interest

The authors declare no conflicts of interest.

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